

CodeType

Programming Typing Practice

CS50x Final Project

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typing engine preview

```
1 for (int i = 0; i < n; i++) {  
2     printf("%i\n", numbers[i]);  
3 }
```

WPM

52.4

Accuracy

96.8%

Mistakes

3

Practice real code · Track progress with SQLite · Explainable in 3 minutes

C

Python

SQL

HTML

JavaScript

Why CodeType?

Most typing sites train prose — programmers need punctuation, indentation, and syntax.

Problem

Ordinary typing practice
sentences · spaces · simple punctuation

Programming typing
brackets · semicolons · operators · exact indentation

Mistakes matter
a single missing character can break real code

Solution

CodeType gives users short, real code snippets and measures speed, accuracy, mistakes, and completion time while saving progress for later review.

Languages supported

C

Python

SQL

HTML

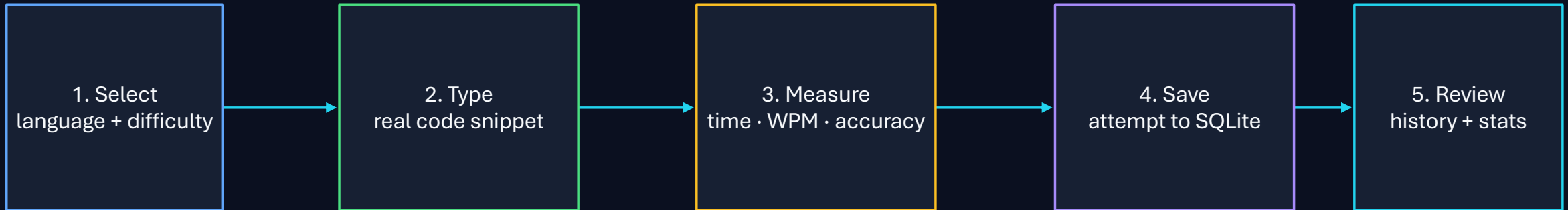
JavaScript

Core idea

**Type code exactly → compare characters →
compute metrics → save attempt**

Demo Flow

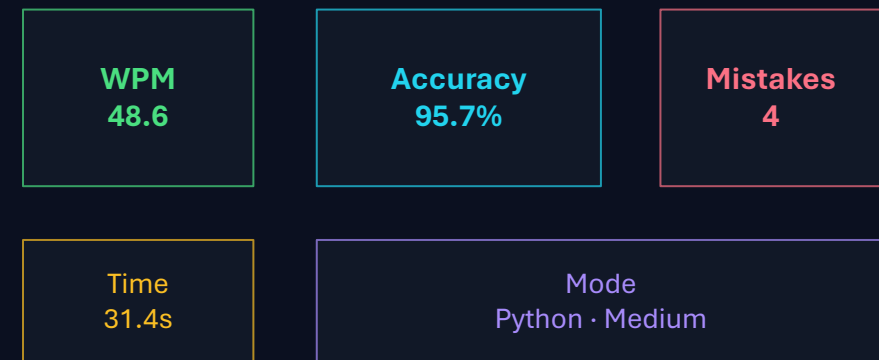
A simple path from language selection to saved history.



In the live demo, I will:

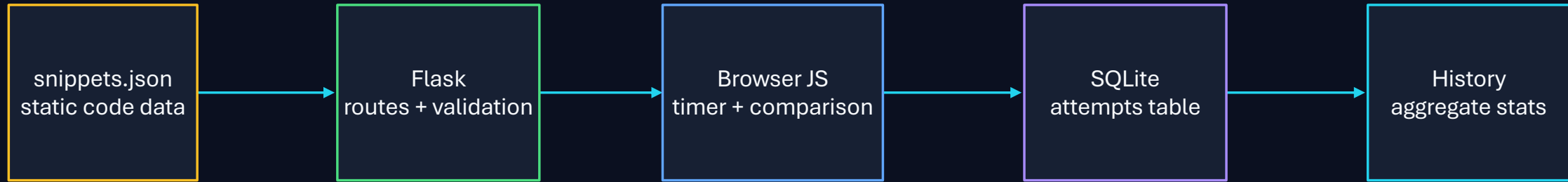
- start a Python or C practice test
- intentionally make and fix a mistake
- finish the snippet to trigger the result panel
- open History to show saved statistics
- clear history to show the database action

Result example



How It Works

Static snippets stay in JSON; dynamic user attempts are stored in SQLite.



Database schema

```
attempts
-----
id INTEGER PRIMARY KEY
snippet_id TEXT
language TEXT
difficulty TEXT
wpm REAL
accuracy REAL
errors INTEGER
duration_seconds REAL
created_at TIMESTAMP
```

SQL used on History page

```
1 SELECT language,
2     COUNT(*) AS tests,
3     MAX(wpm) AS best_wpm,
4     AVG(wpm) AS avg_wpm,
5     AVG(accuracy) AS
avg_accuracy
6 FROM attempts
7 GROUP BY language
8 ORDER BY language;
```

Validation

The backend checks supported language, difficulty, snippet ID, WPM, accuracy, errors, and duration before saving. SQL inserts use parameterized placeholders.

CS50 connection: Python · SQL · Flask · JavaScript · HTML/CSS

What I Built & Learned

A compact CS50 project that combines web programming, SQL, and client-side algorithms.

Python / Flask

server routes, validation, JSON responses

SQLite / SQL

attempt storage, history, aggregate stats

JavaScript

timer, key handling, live metrics, fetch API

HTML / CSS

responsive UI and code-focused typing interface

Algorithmic logic

character comparison, WPM, accuracy, completion

Final message

CodeType is small by design, but complete: users can practice, receive meaningful feedback, and track progress over time.

Future improvements

user profiles · custom snippets · more languages · performance charts · keyboard heatmaps

Thank you!